

Project: Charles Williams
Location: Statham, GA
Description: Williams - Slab

Date: April 16, 2018
By: TJH
House Faces: Worst NorthWest

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals, 2001 Edition, Chapter 28. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Home Diagnostic Solutions.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,945 ft ²
Volume	26,505 ft ³
Building Tightness	Tight
Percentage Glass	10 %

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice Total kWh	12442
15 SEER Comparison kWh	11574
Standard Electric kWh	21060
Standard Elec. Cooling kWh	5709
Gas Heat (Therms)	1036

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.033	Attic	2x6	R-30 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.5	Exterior Door		R-2	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.033	Flat Ceiling		R-30 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.34	Blinds/Draperies	SHGC 0.27
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		32456 Btu/hr		43597 Btu/hr	
Minimum Nominal Capacity		3.61 Tons		12.77 kW	
Load Density		817 SF/Ton		4.3 W/SF	
Floor Area Served		2945.0 SF		2945.0 SF	
System Size		4.0 Tons		15.0 kW	
Leaving Air Temperature		57.2 deg F		99.6 deg F	
Installed Capacity		736 SF/Ton		5.1 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1				
Size	4.00	Tons		
Airflow	1600	CFM		
Heat kW	15	kW		
First Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Entry	63	91	77	
Bed (Front) & WIC	138	148	143	
Bed (Front Left)	132	144	138	
Bed (Front Left) WIC	10	16	13	
Shared Bath (Left)	19	29	24	
Bed (Left)	87	108	97	
Hallway	36	45	41	
Bath (Hall)	13	16	14	
Laundry	65	31	48	
Living	231	148	189	
Breakfast	160	186	173	
Kitchen	253	132	193	
Pantry	24	29	26	
Powder	18	18	18	

[illegible]

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Level(s):

First Floor

Square Feet

2052

Volume

18468

SF/Ton

684

SEER

14

Typical Ceiling Height

9

Cooling Size

3

RC Cooling kWh

3378

3153

Cooling kWh

4329

Heating Size

3

RC Heating kWh

5778

5366

Heating kWh

10772

Kneewall Height

1

Best Case

3

Total RC kWh

9156

8518

Heating Therms

727

14 SEER

15 SEER

COOLING LOAD CALCULATION					Ceiling Height	9		9		9		9		9		9		9		9		9		9		9		9		9		9					
Components				CLTD (°F)	U-value (Btu/hr-sq-ft)	Cooling Load Factor		Entire House		Entry		Bed (Front) & WIC		Bed (Front Left)		Bed (Front Left) WIC		Shared Bath (Left)		Bed (Left)		Hallway		Bath (Hall)		Laundry		Living		Breakfast		Kitchen		Pantry		Powder	
Type	Description					Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1	SouthWest	Right	(E)		31 Btu/hr-sf	36	1120		-		-		-		-		-		-		-		-		-		-	36	1,120		-		-		-	
	G1	West		(SE)		35 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	NorthWest	Front	(S)		22 Btu/hr-sf	72	1551		-	36	776	36	776		-		-		-		-		-		-		-		-		-		-		-	
	G1	North		(SW)		7 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	NorthEast	Left	(W)		9 Btu/hr-sf	18	154		-		-		-		-		-		-	18	154		-		-		-		-		-		-		-	
	G1	East		(NW)		15 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	SouthEast	Back	(N)		17 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	South		(NE)		18 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	Sky Light				55 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
Door	G1	Shaded				7 Btu/hr-sf	62	410	11.07	73		-		-		-		-		-		-		-		-		-	51	337		-		-		-	
	D1	Door	21	0.50		10.50 Btu/hr-sf	25	263	20	210		-		-		-		-		-		-		-		-		-	5	53		-		-		-	
	D2	Door	30	0.50		15.00 Btu/hr-sf	18	267	17.81	267		-		-		-		-		-		-		-		-		-		-		-		-		-	
Above Grade Walls	W1	SouthWest	21	0.08		1.62 Btu/hr-sf	593	959		-	152	245	9	15		-		-		-		-		-		-		-	99	160	273	442	60	97		-	
	W1	West	23	0.08		1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	NorthWest	19	0.08		1.46 Btu/hr-sf	214	314	46	68	82	119	87	127		-		-		-		-		-		-		-		-		-		-		-	
	W1	North	13	0.08		1.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	NorthEast	19	0.08		1.46 Btu/hr-sf	420	615		-		-	162	237	43	63	53	78	162	237		-		-		-		-		-		-		-		-	
	W1	East	23	0.08		1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	SouthEast	21	0.08		1.62 Btu/hr-sf	116	187		-		-		-		-		-		-		-		-		-		-	116	187		-		-		-	
	W1	South	16	0.08		1.23 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	SouthWest	21	0.08		1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
Basement Walls	W1	West	23	0.08		1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	NorthWest	19	0.08		1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	North	13	0.08		1.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	NorthEast	19	0.08		1.46 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	East	23	0.08		1.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	SouthEast	21	0.08		1.62 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	South	16	0.08		1.23 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W2	Grade	0	0.07		0.00 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W1	Partition	10	0.08		0.77 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
Other	W1	Garage	30	0.08		2.31 Btu/hr-sf	280	648	87	201		-		-		-		-		-		-		-		-	63	144		-		-	66	152	65	150	
	W3	Attic	55	0.03		1.82 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	W3	Kneewall	55	0.03		1.82 Btu/hr-sf	91	165		-		-		-		-		-		-		-		-	91	165		-		-		-		-		-	
Floor	R1	Ceiling	55	0.03		1.82 Btu/hr-sf	1627	2953	74	134	246	446	220	399	26	47	60	109	204	370	141	256	50	91	74	134	147	267	233	423	73	132	40	73	39	71	
	R2	Vaulted	55	0.05		2.92 Btu/hr-sf	601	1752		-		-		-		-		-		-		-		-		-	259	1,068		-	166	684		-		-	
	F1	Crawl/Uncon.	11	0.05		0.58 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	F1	Garage	30	0.05		1.59 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
Internal	F2	Slab	0	0.10		0.00 Btu/hr-sf	2052	0	74	-	246	-	220	-	26	60	-	204	-	141	-	50	-	74	-	406	-	233	-	239	-	40	-	39	-		
	People					255 Btu/hr	4	1020		-	1	255	1	255		-	1	255		-		-		-	1	255		-		-		-		-		-	
Floor	Lights (W)					3.412 Btu/hr-W	1028	3508	37	126	123	420	110	375	13	44	30	102	102	348	71	242	25	85	37	126	203	693	117	399	120	409	20	68	20	68	
	Appliances (W)					3.412 Btu/hr-W	1400	4777		-		-		-		-		-		-		-		-	200	682	400	1,365		-	800	2,730		-		-	
	Volume					CF	18468		666		2214		1980		234		540		1836		1269		450		666		3654		2097		2151		360		351		
	Infiltration (Tight)	0.35	ACH		0.13 Btu/hr-sf	2052	2370	74	85	246	284	220	254	26	30	60	69	204	236	141	163	50	58	74	85	406	469	233	269	239	276	40	46	39	45		
	Subtotal (Btu/hr)						23031		1166		2545		2437		185		359		1600		661		234		1193		4260		2947		4674		436		334		
	Safety Factor				10 %		2303																														
	Sensible Cooling Load (Btu/hr)						25334																														
	Nominal Tons Required (75% sens. cap.)						2.81																														
	System Selection	3 Tons		Space CFM			1200		61		133		127		10		19		83		34		12		62		222		154		244		23		17		
	HEATING LOAD CALCULATION					Entire House		Entry		Bed (Front) & WIC		Bed (Front Left)		Bed (Front Left) WIC		Shared Bath (Left)		Bed (Left)		Hallway		Bath (Hall)		Laundry		Living		Breakfast		Kitchen		Pantry		Powder			
Type	Description	AT (°F)	U-Value (Btu/hr-sq-ft)		Heating Load Factor	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr		
Glass	G2	54	0.34		18.4 Btu/hr-sf	188	3453	11.07	203	36	661	36	661	0	-	0	-	18	330	0	-	0	-	0	-	0	-	87	1,597	0	-	0	-	0	-		
	G1	54	0.00		0.0 Btu/hr-sf	0	0		-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	G3	54	0.00		0.0 Btu/hr-sf	0	0		-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Door	D1	54	0.50		27.0 Btu/hr-sf	25																															

Project:	Charles Williams	Date:	April 16, 2018
Location:	Statham, GA	By:	TJH
Description:	Williams - Slab	Level(s):	Master Suite
Square Feet 893 Volume 8037 SF/Ton 893		SEER	14

Typical Ceiling Height	9	Cooling Size	1	RC Cooling kWh	950	886	Cooling kWh	1380
		Heating Size	1.5	RC Heating kWh	2336	2169	Heating kWh	4579
Kneewall Height	1	Best Case	1	Total RC kWh	3286	3056	Heating Therms	309
				14 SEER 15 SEER				

COOLING LOAD CALCULATION					Ceiling Height		9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
Components					Entire House		Master Bed		Master Bath		Master WIC (Small)		Master WIC (Large)									
Type	Description	CLTD (°F)	U-Value (Btu/hr-sq-ft)	Cooling Load Factor	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1 SouthWest	Right (E)		31 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 West	(SE)		35 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 NorthWest	Front (S)		22 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 North	(SW)		7 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 NorthEast	Left (W)		9 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 East	(NW)		15 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 SouthEast	Back (N)		17 Btu/hr-sf	52	858	36	594	16	264		-		-		-		-		-		-
	G1 South	(NE)		18 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	G1 Sky Light			55 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
Door	G1 Shaded			7 Btu/hr-sf	13	88		-	13.36	88		-		-		-		-		-		-
	D1 Door	21	0.50	10.50 Btu/hr-sf	4	47		-	4.452	47		-		-		-		-		-		-
Above Grade Walls	D2 Door	45	0.50	22.50 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	83	134	-	-	83	134	-	-	-	-	-	-	-	-	-	-	-	-
	W1 West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	277	405	-	-	277	405	-	-	-	-	-	-	-	-	-	-	-	-
	W1 East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	316	510	116	187	112	180	-	88	143	-	-	-	-	-	-	-	-	-
	W1 South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Basement Walls	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 Grade	0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	W1 Partition	10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1 Garage	30	0.08	2.31 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W3 Attic	55	0.03	1.82 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W3 Kneewall	55	0.03	1.82 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Roof	R1 Ceiling	55	0.03	1.82 Btu/hr-sf	893	1621	420	762	296	537	56	102	121	220		-		-		-		-
	R2 Vaulted	55	0.05	2.92 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	F1 Crawl/Uncon.	11	0.05	0.58 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	F1 Garage	30	0.05	1.59 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	F2 Slab	0	0.10	0.00 Btu/hr-sf	893	0	420	-	296	-	56	-	121	-	-	-	-	-	-	-	-	-
Internal	People			255 Btu/hr	1	255	1	255								-		-		-		-
	Lights (W)			3.412 Btu/hr-W	447	1525	210	717	148	505	28	96	61	208	-	-	-	-	-	-	-	-
	Appliances (W)			3.412 Btu/hr-W	0	0									-	-	-	-	-	-	-	-
	Volume			CF	8037		3780		2664		504		1089		0	-	0	-	0	-	0	-
	Infiltration (Tight)	0.35 ACH		0.13 Btu/hr-sf	893	1031	420	485	296	342	56	65	121	140	-	-	-	-	-	-	-	-
	Subtotal (Btu/hr)					6475		3405		2098		262		710	0	0	0	0	0	0	0	0
	Safety Factor			10 %		647																
	Sensible Cooling Load (Btu/hr)					7122																
	Nominal Tons Required (75% sens. cap.)					0.79																
	System Selection	1 Tons	Space CFM		400		210		130		16		44		-	-	-	-	-	-	-	-
HEATING LOAD CALCULATION					Entire House		Master Bed		Master Bath		Master WIC		Master WIC		0		0		0		0	
Type	Description	AT (°F)	U-Value (Btu/hr-sq-ft)	Heating Load Factor	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1	54	0.34	18.4 Btu/hr-sf	65	1200	36	661	29.36	539	0	-	0	-	0	-	0	-	0	-	0	-
	G2	54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	G3	54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Walls	D1	54	0.50	27.0 Btu/hr-sf	4	120	0	-	4.452	120	0	-	0	-	0	-	0	-	0	-	0	-
	D2	54	0.50	27.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1	54	0.08	4.2 Btu/hr-sf	675	2807	392.4	1,632	194.5	809	0	-	88.3	367	0	-	0	-	0	-	0	-
	W2	54	0.07	3.6 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W3	54	0.03	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1 Partition	27	0.08	2.1 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W2 Grade	27	0.07	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Roof	R1	54	0.03	1.8 Btu/hr-sf	893	1591	420	748	296	527	56	100	121	216	0	-	0	-	0	-	0	-
	R2	54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F1	54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F2 Slab	27	0.10	2.7 Btu/hr-sf	893	2411	420	1,134	296	799	56	151	121	327	0	-	0	-	0	-	0	-
	Infiltration (Tight)	0.35 ACH		0.35 Btu/hr-sf	893	2785	420	1,310	296	923	56	175	121	377	-	-	-	-	-	-	-	-
	Subtotal (Btu/hr)					10915		5485		3718		426		1287	0	0	0	0	0	0	0	0
	Safety Factor			15 %		1637																
	Heat Loss (Btu/hr)					12552																
	Minimum Heater Size (kW)					3.68																
	System Selection	5 kW	Htg. CFM		400		201		136		16		47		-	-	-	-	-	-	-	-